Rohith Pallamreddy

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Education

Georgia Institute of Technology — Atlanta, GA
Bachelor of Science in Computer Science, GPA: 4.0 — *May 2027*

Experience

Incoming at General Motors Cruise (AV Platform OS and CPU team) — Warren, MI
Autonomous Vehicle Software Engineer Intern — *Summer 2026*

- Will develop and **optimize embedded software platform** components including drivers, operating system, BSP, and software stack for high-performance SoCs powering ADAS and **autonomous driving applications with C++ and RTOS principles.**
- Will work on real-time embedded platform solutions to improve performance, reliability, and **scalability of the core software stack** for production autonomous vehicle systems.

Resilient and Equitable Mobility Analytics and Planning @ Georgia Tech — Atlanta, GA
Machine Learning Researcher (Mapping and Mobility) — *Jan 2025 – Present*

- Generated spatiotemporal processing pipelines to efficiently compress and extract **70+ GB** of data into **<1 GB** for high-performance mobility analysis
- Architected **algorithmic trajectory reconstruction** pipelines using Kalman filters and ML, achieving **96.3% accuracy**, providing applications for high-fidelity **Autonomous Vehicle path planning** and localization
- Generated predictive Motif pipelines with **graph-based models** to identify human decision-making features with **84% accuracy**, providing applications for **human-like routing in Autonomous Vehicles** to increase passenger comfort and the perception of a **safe, human-driven experience**

Perception and Simulation Engineer (Microclimate) — *Aug 2025 - Present*

- Processed **500+ GB of LiDAR datasets** to recreate 3-dimensional terrain, building, and canopy information for downstream environmental simulation
- Utilized LiDAR derived features to implement SOLWEIG to **simulate and predict** perceived microclimate (RMSE $< 1.5°C$), enabling **thermal-aware routing** to mitigate **Autonomous Vehicle hardware overheating**
- Engineered and debugged alternative SOLWEIG pipeline to simulate and predict perceived temperature with a **93% faster runtime**, no spatial compression, and **99.3% accuracy**

Projects

LADEE Lunar Dust Risk Mapping & Path Optimization — Nov 2025 – Dec 2025

- Developed a **high-fidelity Monte Carlo simulation** using **PyBullet** to model 10^5+ dust particles, implementing a danger coefficient that reduced sensor interference risks by 62%.
- Optimized graph-based trajectory planning via **Dijkstra's Algorithm**, improving path safety metrics by 42% across 200+ simulated lunar terrain edge cases.

Motorcycle HIL Simulator — Aug 2025 – Dec 2025

- Developed a Hardware-in-the-Loop (**HIL**) system integrating Arduino-based IMU sensors with a **custom UE4 physics model** to achieve 98% fidelity in real-time lean angle replication.
- Implemented a **Kalman Filter** to fuse noisy accelerometer and gyroscope data, reducing signal variance by 62% and ensuring stable $< 10ms$ latency for high-speed simulation dynamics.

CounterPunch (Computer Vision Defense Analyzer) — May 2025 – Sept 2025

- Architected a real-time perception system using **MediaPipe** to map 3D "strike zones," identifying defensive guard vulnerabilities by calculating spatial coverage gaps in the boxer's stance with **74% accuracy.**
- Engineered a predictive heuristic engine with **90%+ accuracy on beginners** to track head-center coordinates, calculating velocity and acceleration vectors to generate a "Predictability Score" for future movement patterns.

Technical Skills

Languages: C++20 (STL, Boost, Templates, RAII, GDB), Python (C-Extensions, Cython), C, Rust, SQL, Bash
Systems & Sim: High-Fidelity Physics (PyBullet, UE4), HIL/SIL Testing, ROS2 (Pub/Sub, ActionLib), Gazebo
Perception & Math: Computer Vision (OpenCV, MediaPipe), State Estimation (Kalman/EKF, SLAM), Eigen, PCL
ML & Optimization: PyTorch (TensorRT, CUDA Kernels), Scikit-Learn, Distributed Training, Latency Optimization
Tooling & Infra: Linux Kernel/Systems, Bazel (Hermetic Builds), Docker, CI/CD, Git, GCP, Valgrind, Perf